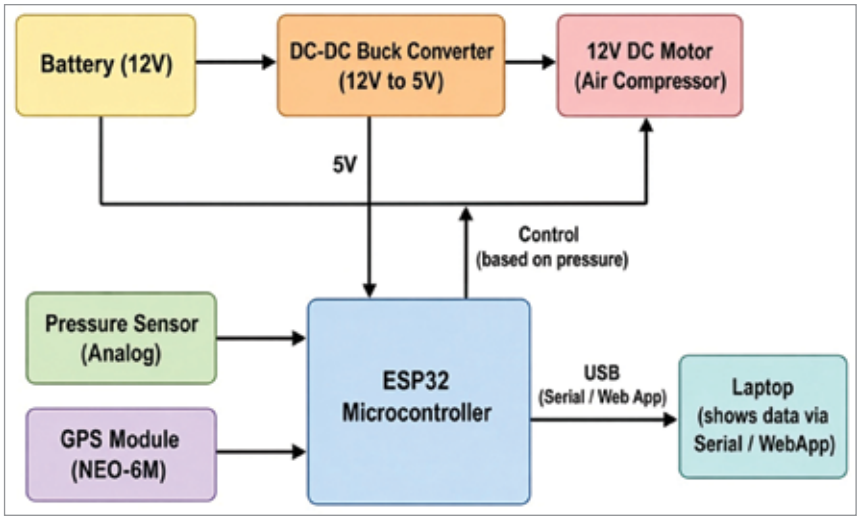


Fig. 2: Block diagram

MOSFET on to run the compressor; at the upper limit, it switches it off, and the loop repeats to hold the required pressure. Meanwhile, the GPS module supplies location data for tracking, and a web dashboard shows live pressure graphs, recorded data, pressure settings, and device status. Fig. 1 shows the author's prototype, while Fig. 2 presents the project's block diagram.

What will you learn?

- How to measure air pressure in real-time and use it to control a compressor automatically.
- How to interface an analogue pressure sensor with the ESP32.
- How to manage power between the battery, motor, and control circuits.
- How to build a web application that displays live pressure graphs, records data, and sets the pressure limit.
- How GPS location data is obtained, and how the automatic cut-off logic works to prevent over- and under-inflation.

READY TO BUILD IT YOURSELF?

Head to **https://efy.co.in/gigh53** for the complete circuit diagram, component list, step-by-step build instructions, troubleshooting tips and demo videos. Everything you need to take this project from idea to working prototype is waiting for you there!

Dr Suganthi S. (left) is a Professor in the ECE Department at K. Ramakrishnan College of Technology, Trichy, with more than 25 years of academic experience and specialisation in microwave engineering.



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